The Journal of Open Source Software

Explabox: Model-Agnostic Machine Learning Transparency & Analysis

Marcel Robeer [1,2], **Michiel Bron** [1,2], **Elize Herrewijnen** [1,2], **Riwish Hoeseni**[2], **and Floris Bex** [1,3]

1 National Police Lab AI, Utrecht University, The Netherlands **2** Netherlands National Police, The Netherlands **3** School of Law, Utrecht University, The Netherlands

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Abhishek Tiwari

Reviewers:

- @hbaniecki
- @JHoelli
- @rachitanaik-lyft

Submitted: 14 March 2025
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

Explabox is an open-source toolkit for transparent and responsible machine learning (ML) development and usage. It promotes explainable, fair, and robust models using a four-step strategy: *explore*, *examine*, *explain* and *expose*. These model-agnostic steps transform complex `ingestibles` (models, data) into interpretable `digestibles`, covering descriptive statistics, performance, local/global explanations, and robustness, security and fairness assessments. Implemented in Python, `Explabox` supports multiple interaction modes, helping developers/testers operationalize explainability, fairness, auditability, and security. The initial release focuses on text data/models, with expansion planned. Code and documentation are available open-source at https://explabox.readthedocs.io.

Statement of need

Responsible and transparent Machine Learning (ML) development and usage has become crucial. High-stakes decisions can significantly impact individuals and society, with severe consequences potentially stemming from biases or errors. For example, the EU AI Act requires high-risk systems to be tested, documented, and assessed before practical application (Edwards, 2022). However, operationalizing transparency (explainable ML) and testing model behavior (fairness, robustness, and security auditing) remains difficult due to the myriad techniques and their learning curves. We devised a four-step analysis strategy—*explore*, *examine*, *explain*, and *expose*—to ensure holistic model transparency and testing. The open-source `Explabox` offers these analyses through well-documented, reproducible steps. Data scientists gain a unified, model-agnostic approach, developed and used in a high-stakes environment, applicable to any text classifier/regressor.

`Explabox` was developed in an organizational environment where models and data are analyzed repeatedly, and where internal and external stakeholders have varying explanatory needs and preferred formats. Several related tools exist, such as `AIX360` (Arya et al., 2019), `alibi explain` (Klaise et al., 2021), `dalex` (Baniecki et al., 2021), `CheckList` (Ribeiro et al., 2020), and `AIF360` (Bellamy et al., 2018). However, these tools may be incompatible with recent Python versions (3.8–3.12), have restricted functionality (testing or explainability focus), lack reproducibility, or lack flexibility communicating results. `Explabox` is an open-source Python toolkit supporting responsible ML development with minimal workflow disruption for practitioners addressing this gap. Its four-step strategy applies to existing data/models and acts as a hub connecting research and best practices.

Explore, Examine, Explain & Expose your ML models

Explabox transforms opaque *ingestibles* into transparent *digestibles* through four types of *analyses* to enhance explainability and aid fairness, robustness, and security audits.

Ingestibles

Ingestibles provide a unified model/data import interface, where layers abstract away access (Figure 1) to allow optimized processing. Explabox uses instancelib (Bron, 2023) for fast model/data encapsulation. The model can be any Python Callable containing a regression or (binary and multi-class) classification model. scikit-learn or onnx models (e.g., PyTorch, TensorFlow/Keras) import directly with optimizations and automatic input/output interpretation. Data can be automatically downloaded, extracted and loaded. Data inputs include NumPy/Pandas/huggingface, raw files (e.g., HDF5, CSV or TSV), and (compressed) file folders. Data can be subdivided into named splits (e.g., train-test-validation), and instance vectors and tokens can be precomputed (and optionally saved) for fast inferencing.

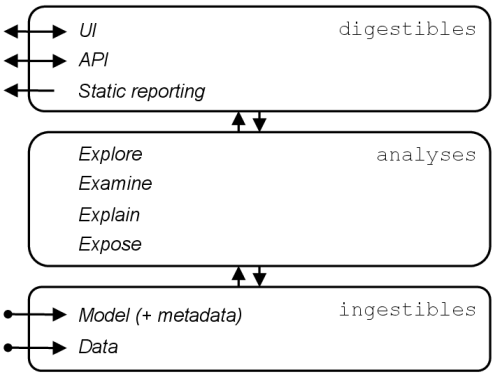

Figure 1: Logical separation of Explabox into layers with interfaces.

Analyses

Explabox turns these *ingestibles* into *digestibles* (transparency-increasing information on ingestibles) through four *analyses* types: **explore**, **examine**, **explain** and **expose**.

Explore allows data slicing/dicing/sorting, and provides descriptive statistics (dataset sizes, label distributions, and text string/token lengths).

Examine shows model performance metrics, summarized in a table or shown graphically, and include computation/interpretation references. For further analysis, **examine** also supports drilling down into (in)correct predictions.

Explain uses model-agnostic techniques (Ribeiro et al., 2016a) to explain model behavior (*global*) and individual predictions (*local*). It summarizes model-labelled data, through prototypes (KMedoids) or prototypes and criticisms (MMDCritic (Kim et al., 2016)), and token distributions (TokenFrequency) and informativeness (TokenInformation). Local explanations use popular techniques: feature attribution scores (LIME (Ribeiro et al., 2016b), KernelSHAP (Lundberg & Lee, 2017)), feature subsets (Anchors (Ribeiro et al., 2018)), local rule-based models (LORE (Guidotti et al., 2018)), and counterfactual/contrastive explanations (FoilTrees (Waa et al., 2018)). Built from generic components separating global and local explanation steps, these methods allow customization and enable scientific advances to be quickly integrated into operational processes (e.g., combine KernelShap sampling with imodels (Singh et al., 2021) surrogate rules). Example configurations, such as LIME with default hyperparameters, ease

DRAFT

adoption. **Explain** is provided by subpackage `text_explainability` (Robeer, 2021a), which doubles as a standalone tool.

Expose gathers sensitivity insights via local/global testing. These insights can be used to, through relevant attributes, assess the *robustness* (e.g., the effect of typos on model performance), *security* (e.g., if inputs containing certain characters crash the model), and *fairness* (e.g., subgroup performance for protected attributes such as country of origin, gender, race or socioeconomic status) of the model. Relevant attributes can either be observed in the current data or generated from user-provided templates (Ribeiro et al., 2020) filled with multi-language data generation (Faraglia & Other Contributors, 2021). These attributes are then either summarized in performance metrics, compared to expected behavior (Ribeiro et al., 2020), or assessed with fairness metrics for classification (Mehrabi et al., 2021) and regression (Agarwal et al., 2019). Like **explain**, **expose** is also made from generic components, which allows users to customize data generation and tests. **Expose** is provided by the `text_sensitivity` subpackage (Robeer, 2021b), which also doubles as a standalone tool.

Digestibles

Digestibles serve stakeholders—such as creators, auditors, applicants, end-users or clients (Tomsett et al., 2018)—via a Jupyter/web UI (Figure 2) (using `plotly` (Plotly Technologies Inc., 2015) visuals), integrated API, and static reporting.

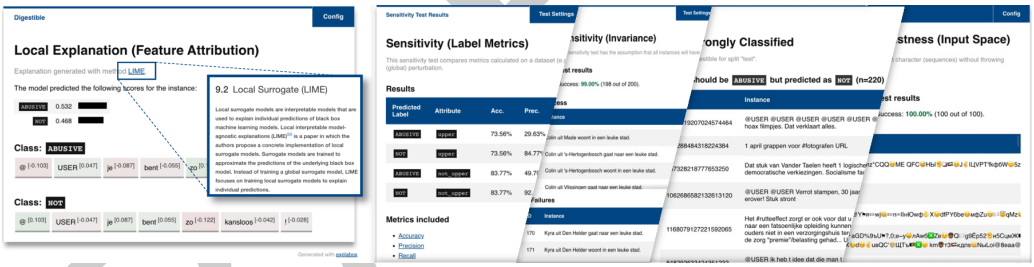

Figure 2: UI elements of the Jupyter Notebook interface for interactive explainability and analyses.

Acknowledgements

Development was supported by the Netherlands National Police. The authors thank contributors within the Police for development, testing, and usage, and participants from ICT.OPEN 2022, the UU NPAI, and UMC Utrecht demos for their valuable feedback.

References

Agarwal, A., Dudik, M., & Wu, Z. S. (2019). Fair Regression: Quantitative Definitions and Reduction-Based Algorithms. In K. Chaudhuri & R. Salakhutdinov (Eds.), *Proceedings of the 36th international conference on machine learning* (Vol. 97, pp. 120–129). PMLR.

Arya, V., Bellamy, R. K. E., Chen, P.-Y., Dhurandhar, A., Hind, M., Hoffman, S. C., Houde, S., Liao, Q. V., Luss, R., Mojsilović, A., Mourad, S., Pedemonte, P., Raghavendra, R., Richards, J., Sattigeri, P., Shanmugam, K., Singh, M., Varshney, K. R., Wei, D., & Zhang, Y. (2019). One Explanation Does Not Fit All: A Toolkit and Taxonomy of AI Explainability Techniques. *arXiv Preprint*. https://doi.org/arXiv.1909.03012

Baniecki, H., Kretowicz, W., Piatyszek, P., Wisniewski, J., & Biecek, P. (2021). dalex: Responsible Machine Learning with Interactive Explainability and Fairness in Python. *Journal of Machine Learning Research*, *22*(214), 1–7. http://jmlr.org/papers/v22/20-1473.html

DRAFT

Bellamy, R. K. E., Dey, K., Hind, M., Hoffman, S. C., Houde, S., Kannan, K., Lohia, P., Martino, J., Mehta, S., Mojsilovic, A., Nagar, S., Ramamurthy, K. N., Richards, J., Saha, D., Sattigeri, P., Singh, M., Varshney, K. R., & Zhang, Y. (2018). AI Fairness 360: An Extensible Toolkit for Detecting, Understanding, and Mitigating Unwanted Algorithmic Bias. *arXiv Preprint*. https://doi.org/arXiv.1810.01943

Bron, M. P. (2023). *Python Package instancelib* (Version 0.5.0). Zenodo. https://doi.org/10.5281/zenodo.8308017

Edwards, L. (2022). The EU AI Act: A summary of its significance and scope. *Ada Lovelace Institute, Expert Explainer Report*. https://www.adalovelaceinstitute.org/wp-content/uploads/2022/04/Expert-explainer-The-EU-AI-Act-11-April-2022.pdf

Faraglia, D., & Other Contributors. (2021). *Python package Faker*. https://github.com/joke2k/faker

Guidotti, R., Monreale, A., Ruggieri, S., Pedreschi, D., Turini, F., & Giannotti, F. (2018). Local Rule-Based Explanations of Black Box Decision Systems. *arXiv Preprint*. https://doi.org/arXiv.1805.10820

Kim, B., Khanna, R., & Koyejo, O. (2016). Examples are not Enough, Learn to Criticize! Criticism for Interpretability. *29th Conference on Neural Information Processing Systems (NIPS 2016)*.

Klaise, J., Looveren, A. V., Vacanti, G., & Coca, A. (2021). Alibi Explain: Algorithms for Explaining Machine Learning Models. *Journal of Machine Learning Research*, *22*(181), 1–7. http://jmlr.org/papers/v22/21-0017.html

Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, 4765–4774.

Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A Survey on Bias and Fairness in Machine Learning. *ACM Computing Surveys (CSUR)*, *54*(6), 1–35. https://doi.org/10.1145/3457607

Plotly Technologies Inc. (2015). *Collaborative data science*. Plotly Technologies Inc. https://plot.ly

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016a). Model-Agnostic Interpretability of Machine Learning. *2016 ICML Workshop on Human Interpretability in Machine Learning (WHI 2016)*, 91–95. https://doi.org/10.1145/2858036.2858529

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016b). "Why Should I Trust You?": Explaining the Predictions of Any Classifier. *22nd ACM SIGKDD International Conference on Knowledge Discovery in Data Mining (KDD'16), Proceedings*, 1135–1144. https://doi.org/10.18653/v1/n16-3020

Ribeiro, M. T., Singh, S., & Guestrin, C. (2018). Anchors: High-Precision Model-Agnostic Explanations. *AAAI Conference on Artificial Intelligence, Proceedings*. https://doi.org/10.1609/aaai.v32i1.11491

Ribeiro, M. T., Wu, T., Guestrin, C., & Singh, S. (2020). Beyond Accuracy: Behavioral Testing of NLP models with CheckList. *Association for Computational Linguistics (ACL)*. https://doi.org/10.18653/v1/2020.acl-main.442

Robeer, M. (2021a). *Python package text_explainability*. https://doi.org/10.5281/zenodo.14192126

Robeer, M. (2021b). *Python package text_sensitivity*. https://doi.org/10.5281/zenodo.14192940

Singh, C., Nasseri, K., Tan, Y. S., Tang, T., & Yu, B. (2021). imodels: A python package for fitting interpretable models. *Journal of Open Source Software*, *6*(61), 3192. https:

DRAFT

//doi.org/10.21105/joss.03192

Tomsett, R., Braines, D., Harborne, D., Preece, A., & Chakraborty, S. (2018). Interpretable to Whom? A Role-based Model for Analyzing Interpretable Machine Learning Systems. *2018 ICML Workshop on Human Interpretability in Machine Learning (WHI 2018)*.

Waa, J. van der, Robeer, M., Diggelen, J. van, Neerincx, M., & Brinkhuis, M. (2018). Contrastive Explanations with Local Foil Trees. *2018 Workshop on Human Interpretability in Machine Learning (WHI 2018)*.

DRAFT